

```
In [2]: import pandas as pd  
import numpy as np  
import matplotlib.pyplot as plt
```

```
In [3]: pd.read_csv(r"C:\Users\HP\Documents\largest financial services companies by revenue.csv")
```

Out[3]:

	Rank	Company	Industry	Revenue in (USD Million)	Net Income in (USD Millions)	Total Assest in (USD Millions)	Headquarters
0	1	Transamerica Corporation	Conglomerate	245510	42521	873	United States
1	2	Ping An Insurance Group	Insurance	191509	20738	1460	China
2	3	ICBC	Banking	182794	45783	5110	China
3	4	China Construction Bank	Banking	172000	39282	4311	China
4	5	Agricultural Bank of China	Banking	153884	31293	4169	China
5	6	China Life Insurance	Insurance	144589	4648	776	China
6	7	Allianz	Insurance	136173	7756	1297	Germany
7	8	Bank of China	Banking	134045	27952	3739	China
8	9	JP Morgan Chase	Banking	129503	29131	3386	United States
9	10	AXA	Insurance	128011	3605	984	France
10	11	Fannie Mae	Investment Services	106437	11805	3985	United States
11	12	Life Insurance Corporation of India	Insurance	102851	4954	630	India
12	13	Generali Group	Insurance	97128	1987	667	Italy
13	14	Bank of America	Banking	93753	17894	2819	United States
14	15	Citigroup	Banking	88839	11047	2260	United States
15	16	People's Insurance Company	Insurance	84290	2903	192	China
16	17	Credit Agricole	Banking	82958	3067	2399	France
17	18	BNP Paribas	Banking	81632	8052	3045	France
18	19	HSBC	Banking	80429	5229	2984	United Kingdom

	Rank	Company	Industry	Revenue in (USD Million)	Net Income in (USD Millions)	Total Assest in (USD Millions)	Headquarters
19	20	Wells Fargo	Banking	80303	3301	1955	United States
20	21	State Farm	Insurance	78898	3738	299	United States
21	22	Nippon Life	Insurance	76984	3127	773	Japan
22	23	Munich Re	Insurance	74074	1379	364	Germany
23	24	Dai-ichi Life	Insurance	73841	3431	574	Japan
24	25	Banco Santander	Banking	73630	9639	1845	Spain
25	26	MetLife	Insurance	67842	5407	795	United States
26	27	Bank of Communications	Banking	67605	11409	1639	China
27	28	Freddie Mac	Investment Services	66228	7326	2627	United States
28	29	Legal and General Group	Insurance	63324	2061	779	United Kingdom
29	30	Brookfield Assest Management	Investment Services	62752	448	343	Canada
30	31	Aviva	Insurance	62579	3588	655	United Kingdom
31	32	China Pacific Insurance	Insurance	61185	3562	271	China
32	33	China Merchants Bank	Banking	60433	14107	1281	China
33	34	Zurich Insurance Group	Insurance	59001	3834	439	Switzerland
34	35	Manulife Financial	Insurance	58840	4377	691	Canada
35	36	Prudential Financial	Insurance	57033	448	940	United States
36	37	Mitsubishi UFJ Financial Group	Banking	56838	7329	3250	Japan

	Rank	Company	Industry	Revenue in (USD Million)	Net Income in (USD Millions)	Total Assest in (USD Millions)	Headquarters
37	38	Prudential	Insurance	55973	2118	516	United Kingdom
38	39	StoneX Group Inc	Investment Services	54139	169	13	United States
39	40	Goldman Sachs	Investment Services	53498	9459	1163	United States
40	41	Industrial Bank	Banking	53313	9655	1209	China
41	42	Shanghai Pudong Development	Banking	52628	8443	1219	China
42	43	Morgan Stanley	Investment Services	52047	10996	1115	United States
43	44	State Bank of India	Banking	51919	3018	662	India
44	45	Tokio Marine Holdings	Insurance	51516	1526	232	Japan
45	46	AIA Group	Insurance	50359	5779	326	China
46	47	China Mensheng Banking	Banking	49076	4972	1065	China
47	48	Power Corporation of Canada	Insurance	48183	1525	493	Canada
48	49	Talanx	Insurance	46788	766	221	Germany
49	50	Lloyds Banking Group	Banking	14592	1882	1183	United Kingdom

```
In [4]: fin_df=pd.read_csv(r"C:\Users\HP\Documents\largest financial services companies by revenue.csv")
```

```
In [5]: fin_df.shape
```

```
Out[5]: (50, 7)
```

In [6]: `fin_df.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50 entries, 0 to 49
Data columns (total 7 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Rank                                  50 non-null     int64
1   Company                             50 non-null     object
2   Industry                             50 non-null     object
3   Revenue in (USD Million)             50 non-null     int64
4   Net Income in (USD Millions)         50 non-null     int64
5   Total Assest in (USD Millions)      50 non-null     int64
6   Headquarters                         50 non-null     object
dtypes: int64(4), object(3)
memory usage: 2.9+ KB
```

In [7]: `fin_df.isnull().sum()`

```
Out[7]: Rank                                0
Company                                0
Industry                              0
Revenue in (USD Million)              0
Net Income in (USD Millions)          0
Total Assest in (USD Millions)        0
Headquarters                          0
dtype: int64
```

In [8]: *# Clean column names for easier handling*
`fin_df.columns = [col.strip().replace(" ", "_").replace("(", "").replace(")", "") for col in fin_df.columns]`

In [9]: *#Companies with the Highest Revenue*
`fin_df.sort_values(by='Revenue_in_USD_Million', ascending=False).head(10)`

Out[9]:

	Rank	Company	Industry	Revenue_in_USD_Million	Net_Income_in_USD_Millions	Total_Assest_in_USD_Millions	Headquarters
0	1	Transamerica Corporation	Conglomerate	245510	42521	873	United States
1	2	Ping An Insurance Group	Insurance	191509	20738	1460	China
2	3	ICBC	Banking	182794	45783	5110	China
3	4	China Construction Bank	Banking	172000	39282	4311	China
4	5	Agricultural Bank of China	Banking	153884	31293	4169	China
5	6	China Life Insurance	Insurance	144589	4648	776	China
6	7	Allianz	Insurance	136173	7756	1297	Germany
7	8	Bank of China	Banking	134045	27952	3739	China
8	9	JP Morgan Chase	Banking	129503	29131	3386	United States
9	10	AXA	Insurance	128011	3605	984	France

-Insights

1. Revenue Leaders

Transamerica Corporation leads the list with \$245.5B in revenue, significantly ahead of the rest.

Chinese banks dominate the top ranks: ICBC, China Construction Bank, Agricultural Bank of China, and Bank of China all appear in the top 10.

The insurance sector is also strongly represented (Ping An, China Life, Allianz, AXA).

2. Net Income Performance

ICBC reports the highest net income at *45.8B*, followed by *Transamerica* (42.5B) and China Construction Bank (\$39.3B).

High revenue does not always equate to high profitability; for example, AXA earns only *3.6B net income* on 128B revenue—suggesting low profit margins.

3. Industry Trends

Banking firms generally show stronger profit margins and higher total assets compared to insurance firms.

The Conglomerate category (Transamerica) may benefit from diversified services but still performs exceptionally well in both revenue and net income.

4. Regional Dominance

China is a powerhouse, hosting five of the top 10 companies, especially in banking.

United States, Germany, and France are also present but with fewer entries.

5. Asset Utilization

Despite having lower total assets compared to the big banks, Transamerica Corporation achieves a very high revenue-to-asset ratio—suggesting high operational efficiency.

Insurance companies tend to have lower asset levels compared to banks but vary widely in profitability.

-Recommendations For Investors: -Consider Chinese banks (ICBC, CCB, ABC, BOC) for their consistent high revenue, asset strength, and profitability.

-Monitor low-margin insurers like AXA for restructuring or cost-optimization strategies before investing.

-For Financial Institutions: Benchmark asset efficiency against Transamerica. High returns with lower asset bases indicate optimized operations worth emulating.

Explore diversification—Transamerica's conglomerate model seems to work effectively.

-For Policymakers & Analysts: China's dominance in the banking sector suggests strong governmental or regulatory support. Study policy environments that foster such financial growth for possible adaptation.

Encourage transparency in profitability metrics across industries to allow better investor and consumer understanding.

```
In [10]: #Correlation Between Revenue and Net Income
fin_df[['Revenue_in_USD_Million', 'Net_Income_in_USD_Millions']].corr()
```

```
Out[10]:
```

	Revenue_in_USD_Million	Net_Income_in_USD_Millions
Revenue_in_USD_Million	1.000000	0.788733
Net_Income_in_USD_Millions	0.788733	1.000000

strong positive correlation ($r = 0.79$) between revenue and net income.

-Insights

1. Strong but Not Perfect Relationship

A correlation of 0.79 indicates that as revenue increases, net income tends to increase as well, but not always proportionally.

There are other factors—like operating efficiency, cost structures, and debt levels—that impact net income beyond revenue alone.

2. Operational Efficiency Matters

Some companies may earn similar revenues but differ greatly in net income, highlighting differences in cost control or strategic investment.

For instance, a company with high operating costs or large write-downs may earn billions in revenue but post relatively weak profits.

3. Useful for Benchmarking

This correlation helps identify underperformers: Companies with high revenue but lower-than-expected net income might need internal review.

Likewise, companies with higher-than-average income per revenue dollar (high profit margin) are often more efficient or innovative.

-Recommendations For Financial Analysts: Use this correlation to flag outliers—companies where net income is disproportionately low or high compared to revenue.

Conduct profitability margin analysis (Net Income ÷ Revenue) across peers to identify operationally efficient firms.

-For Executives: Focus not just on top-line growth (revenue) but on cost optimization and strategic spending to boost bottom-line performance.

Investigate competitors with similar revenues but higher net income—what are they doing differently?

-For Investors: Revenue growth is a good signal—but not the whole story. Look for companies that turn revenue into sustainable profit.

Diversify into firms with both high revenue and high margins for lower risk and long-term returns.

```
In [11]: #Industry with Highest Average Revenue
fin_df.groupby('Industry')['Revenue_in_USD_Million'].mean().sort_values(ascending=False)
```

```
Out[11]: Industry
Conglomerate      245510.000000
Banking           88008.700000
Insurance         81346.565217
Investment Services 65850.166667
Name: Revenue_in_USD_Million, dtype: float64
```

Insights:

-The conglomerate category far exceeds other industries in average revenue — nearly 3x higher than banking and almost 4x higher than insurance.

-This likely reflects diversified services, cross-sector reach, and larger market footprints.

-Banks, on average, generate more revenue than insurance companies, though the gap is not extreme.

-Banking's edge may come from broader service offerings (lending, corporate banking, trading) and scale of operations.

-Investment services firms have the lowest average revenue

-Recommendations

-Conglomerate structure offers scale—consider vertical or horizontal integration to diversify revenue sources.

-Explore opportunities to bundle services across industries (e.g., insurance + investment).

-For Startups competing directly with conglomerates may be difficult—focus on niche markets or specialization.

-For Startups consider starting in investment services, where barriers to entry may be lower, and scaling strategically.

-For Investor diversify across industries—conglomerates for scale, banks for stability, and insurance/investment for risk-adjusted performance.

-For Investor evaluate if the high revenue of conglomerates translates into high margins, not just volume.

```
In [12]: # Industry with Highest Total Assets
fin_df.groupby('Industry')['Total_Assest_in_USD_Millions'].mean().sort_values(ascending=False)
```

```
Out[12]: Industry
Banking          2476.500000
Investment Services 1541.000000
Conglomerate       873.000000
Insurance         624.956522
Name: Total_Assest_in_USD_Millions, dtype: float64
```

-Insights: -Banking stands out with the highest total assets at \$2,476.50 million. This indicates that the banking sector has significant financial resources, possibly reflecting a well-established and stable industry, with strong market penetration and liquidity.

-Investment Services follows with \$1,541.00 million. This suggests that investment services are a key player in the financial sector, albeit with relatively lower assets compared to banking. It could imply that the industry is capital-intensive but not as large as banking.

-Conglomerates have \$873.00 million in assets. This shows a moderate asset base, which likely reflects diversified business operations across several sectors. Conglomerates can be seen as a stable industry but may be more vulnerable to fluctuations in specific sectors they are involved in.

-Insurance has the lowest total assets at \$624.96 million. Despite its critical role in risk management and financial protection, it appears smaller in terms of assets compared to the other sectors. This could be due to lower capital requirements in comparison to industries like banking or investment services.

-Recommendation The banking sector should leverage its large asset base to expand into digital services and emerging markets, while investment services should diversify into alternative assets to capitalize on new trends. Conglomerates should focus on operational efficiency and consider strategic mergers, while insurance companies should expand market reach through innovative offerings and partnerships.

```
In [13]: #Countries with the Most Companies
fin_df['Headquarters'].value_counts()
```

```
Out[13]: Headquarters
China          14
United States  13
United Kingdom  5
Japan           4
Germany         3
France          3
Canada          3
India           2
Italy           1
Spain           1
Switzerland     1
Name: count, dtype: int64
```

Insights -China and the United States dominate the list, with 14 and 13 company headquarters respectively, indicating their strong global presence and economic influence in the industry.

-The United Kingdom, Japan, and several European countries (Germany, France, Switzerland, etc.) contribute a moderate number of headquarters, reflecting a broad but less concentrated global distribution.

-Emerging economies like India are beginning to show up with multiple entries, suggesting growing influence and participation in the global financial or corporate landscape.

-European countries collectively (UK, Germany, France, Italy, Spain, Switzerland) account for a significant portion, highlighting Europe's continued importance as a hub for major corporations.

```
In [14]: #Average Revenue, Income, and Assets by Country
fin_df.groupby('Headquarters')[['Revenue_in_USD_Million', 'Net_Income_in_USD_Millions', 'Total_Assest_in_USD_Millions']].mean()
```

Out[14]:

	Revenue_in_USD_Million	Net_Income_in_USD_Millions	Total_Assest_in_USD_Millions
Headquarters			
Canada	56591.666667	2116.666667	509.000000
China	104122.142857	16466.142857	1911.928571
France	97533.666667	4908.000000	2142.666667
Germany	85678.333333	3300.333333	627.333333
India	77385.000000	3986.000000	646.000000
Italy	97128.000000	1987.000000	667.000000
Japan	64794.750000	3853.250000	1207.250000
Spain	73630.000000	9639.000000	1845.000000
Switzerland	59001.000000	3834.000000	439.000000
United Kingdom	55379.400000	2975.600000	1223.400000
United States	90310.000000	11787.846154	1710.000000

-Insights: -China leads in net income (\$16.47B) and has one of the highest revenues and asset bases, indicating strong profitability and financial scale.

-France and the United States show balanced financial profiles with high revenues and relatively high net incomes and assets, reflecting mature, diversified economies.

-Spain stands out with a high net income (\$9.64B) relative to its revenue and assets—suggesting strong operational efficiency or high-margin business models.

-India and Italy have strong revenues but lower asset bases, which may indicate a leaner operational structure or higher efficiency.

-Canada and Switzerland show modest figures across all metrics, suggesting either smaller markets or more specialized operations.

-Recommendations: China, France, and the U.S. should continue leveraging their scale to dominate global markets while investing in innovation to sustain profitability. Countries like India, Italy, and Canada should focus on optimizing asset utilization and expanding high-margin segments to boost net income.

```
In [15]: #Calculate the profit Margin  
fin_df['Profit_Margin'] = fin_df['Net_Income_in_USD_Millions'] / fin_df['Revenue_in_USD_Million']  
  
In [16]: print(fin_df['Profit_Margin'])
```

0	0.173195
1	0.108287
2	0.250462
3	0.228384
4	0.203354
5	0.032146
6	0.056957
7	0.208527
8	0.224945
9	0.028162
10	0.110911
11	0.048167
12	0.020458
13	0.190863
14	0.124349
15	0.034441
16	0.036971
17	0.098638
18	0.065014
19	0.041107
20	0.047378
21	0.040619
22	0.018617
23	0.046465
24	0.130911
25	0.079700
26	0.168760
27	0.110618
28	0.032547
29	0.007139
30	0.057336
31	0.058217
32	0.233432
33	0.064982
34	0.074388
35	0.007855
36	0.128945
37	0.037840
38	0.003122
39	0.176810
40	0.181100

```

41    0.160428
42    0.211271
43    0.058129
44    0.029622
45    0.114756
46    0.101312
47    0.031650
48    0.016372
49    0.128975

```

Name: Profit_Margin, dtype: float64

```

In [17]: #Industry with Highest Profit Margins
fin_df.groupby('Industry')['Profit_Margin'].mean().sort_values(ascending=False)

```

```

Out[17]: Industry
Conglomerate      0.173195
Banking            0.148230
Investment Services 0.103312
Insurance          0.047259
Name: Profit_Margin, dtype: float64

```

Insight: Conglomerates have the highest average profit margin (17.32%), suggesting they operate with high efficiency or benefit from diversified income streams. Banking (14.82%) also shows solid profitability, likely due to stable revenue sources and economies of scale. Investment Services (10.33%) trail behind, possibly due to market volatility or fee compression. Insurance (4.73%) has the lowest profit margin, indicating tighter margins likely caused by high claim payouts, regulation, or competition.

Recommendation: Conglomerates and banks should reinvest profits to fuel innovation and strategic expansion, reinforcing their competitive edge. Investment services and insurance firms should focus on cost optimization, digital transformation, and customer-centric models to improve operational efficiency and margins.

```

In [18]: #Country-wise Profitability
fin_df.groupby('Headquarters')['Profit_Margin'].mean().sort_values(ascending=False)

```

```
Out[18]: Headquarters
China      0.148829
Spain      0.130911
United States 0.115548
Switzerland 0.064982
United Kingdom 0.064342
Japan      0.061413
France     0.054590
India      0.053148
Canada     0.037726
Germany    0.030648
Italy      0.020458
Name: Profit_Margin, dtype: float64
```

-Insight: China (14.88%), Spain (13.09%), and the United States (11.55%) lead in average profit margins, indicating stronger operational efficiency and better cost-to-revenue ratios in companies headquartered there. European countries like Germany (3.06%), Italy (2.05%), and Canada (3.77%) show relatively low margins, which may point to higher operating costs, regulatory burdens, or less profitable business models.

-Recommendation: Countries with high profit margins—like China, Spain, and the U.S.—should focus on sustaining their edge through innovation, cost management, and international expansion. Lower-margin regions such as Germany, Italy, and Canada should prioritize efficiency improvements, strategic restructuring, or digital investments to enhance profitability.

```
In [19]: #Calculate the revenue per asset
fin_df['Revenue_per_Asset'] = fin_df['Revenue_in_USD_Million'] / fin_df['Total_Assest_in_USD_Millions']
```

```
In [20]: print(fin_df['Revenue_per_Asset'])
```


0	281.225659
1	131.170548
2	35.771820
3	39.897936
4	36.911490
5	186.326031
6	104.990748
7	35.850495
8	38.246604
9	130.092480
10	26.709410
11	163.255556
12	145.619190
13	33.257538
14	39.309292
15	439.010417
16	34.580242
17	26.808539
18	26.953418
19	41.075703
20	263.872910
21	99.591203
22	203.500000
23	128.642857
24	39.907859
25	85.335849
26	41.247712
27	25.210506
28	81.288832
29	182.950437
30	95.540458
31	225.774908
32	47.176425
33	134.398633
34	85.151954
35	60.673404
36	17.488615
37	108.474806
38	4164.538462
39	46.000000
40	44.096774

```
41      43.173093
42      46.678924
43      78.427492
44     222.051724
45     154.475460
46      46.080751
47      97.734280
48     211.710407
49      12.334742
```

Name: Revenue_per_Asset, dtype: float64

```
In [21]: # Calculate the income per Asset
fin_df['Income_per_Asset'] = fin_df['Net_Income_in_USD_Millions'] / fin_df['Total_Assest_in_USD_Millions']
```

```
In [22]: print(fin_df['Income_per_Asset'])
```

0	48.706758
1	14.204110
2	8.959491
3	9.112039
4	7.506117
5	5.989691
6	5.979954
7	7.475796
8	8.603367
9	3.663618
10	2.962359
11	7.863492
12	2.979010
13	6.347641
14	4.888053
15	15.119792
16	1.278449
17	2.644335
18	1.752346
19	1.688491
20	12.501672
21	4.045278
22	3.788462
23	5.977352
24	5.224390
25	6.801258
26	6.960952
27	2.788732
28	2.645700
29	1.306122
30	5.477863
31	13.143911
32	11.012490
33	8.733485
34	6.334298
35	0.476596
36	2.255077
37	4.104651
38	13.000000
39	8.133276
40	7.985939

```
41    6.926169
42    9.861883
43    4.558912
44    6.577586
45   17.726994
46    4.668545
47    3.093306
48    3.466063
49    1.590871
```

Name: Income_per_Asset, dtype: float64

```
In [23]: #Highest Revenue per Unit of Total Assets
fin_df.sort_values(by='Revenue_per_Asset', ascending=False).head(10)
```

Out[23]:

	Rank	Company	Industry	Revenue_in_USD_Million	Net_Income_in_USD_Millions	Total_Assest_in_USD_Millions	Headquarters	P
38	39	StoneX Group Inc	Investment Services	54139	169	13	United States	
15	16	People's Insurance Company	Insurance	84290	2903	192	China	
0	1	Transamerica Corporation	Conglomerate	245510	42521	873	United States	
20	21	State Farm	Insurance	78898	3738	299	United States	
31	32	China Pacific Insurance	Insurance	61185	3562	271	China	
44	45	Tokio Marine Holdings	Insurance	51516	1526	232	Japan	
48	49	Talanx	Insurance	46788	766	221	Germany	
22	23	Munich Re	Insurance	74074	1379	364	Germany	
5	6	China Life Insurance	Insurance	144589	4648	776	China	
29	30	Brookfield Assest Management	Investment Services	62752	448	343	Canada	



-Insights: 1-Transamerica Corporation (Conglomerate) stands out with a remarkably high profit margin (17.32%) and income per asset (48.71), showcasing exceptional efficiency and profitability—well above industry peers.

2-StoneX Group Inc has extremely high revenue per asset (4164.54) but very low profit margin (0.31%), suggesting high operational volume with minimal profitability—possibly due to thin margins or high costs.

3-Among insurance companies, People's Insurance Company and China Pacific Insurance show relatively strong profit margins (3.44% and 5.82%) and decent asset efficiency, indicating healthy financial performance.

4-Brookfield Asset Management shows low profit margin (0.71%) and very low income per asset (1.31), suggesting underperformance relative to peers and potential inefficiencies in asset utilization.

5-German insurers (Talanx and Munich Re) reflect low profitability with income per asset ratios below 4, implying limited returns on their asset base.

-Recommendations : Transamerica should continue leveraging its highly efficient model by scaling operations and entering new markets, while StoneX Group must reassess cost structures or pricing models to improve profitability. Insurance and investment firms with low margins, such as Brookfield, Munich Re, and Talanx, should focus on enhancing operational efficiency and explore higher-margin service lines or technological investments to boost return on assets.

```
In [24]: #Highest Net Income per Unit of Total Assets  
fin_df.sort_values(by='Income_per_Asset', ascending=False).head(10)
```

Out[24]:

	Rank	Company	Industry	Revenue_in_USD_Million	Net_Income_in_USD_Millions	Total_Assest_in_USD_Millions	Headquarters	P
0	1	Transamerica Corporation	Conglomerate	245510	42521	873	United States	
45	46	AIA Group	Insurance	50359	5779	326	China	
15	16	People's Insurance Company	Insurance	84290	2903	192	China	
1	2	Ping An Insurance Group	Insurance	191509	20738	1460	China	
31	32	China Pacific Insurance	Insurance	61185	3562	271	China	
38	39	StoneX Group Inc	Investment Services	54139	169	13	United States	
20	21	State Farm	Insurance	78898	3738	299	United States	
32	33	China Merchants Bank	Banking	60433	14107	1281	China	
42	43	Morgan Stanley	Investment Services	52047	10996	1115	United States	
3	4	China Construction Bank	Banking	172000	39282	4311	China	



-Insights: -Transamerica Corporation (Conglomerate, USA) leads with a profit margin of 17.32% and an impressive income per asset of 48.71, indicating exceptional efficiency in converting assets into profits.

-AIA Group (Insurance, China) showcases a strong profit margin of 11.48% and income per asset of 17.73, reflecting robust profitability and effective asset utilization.

-Ping An Insurance Group (Insurance, China) maintains a solid profit margin of 10.83% and income per asset of 14.20, suggesting consistent performance in the insurance sector.

-StoneX Group Inc (Investment Services, USA) exhibits a high revenue per asset of 4164.54 but a low profit margin of 0.31%, indicating high operational volume with minimal profitability.

-China Merchants Bank (Banking, China) reports a remarkable profit margin of 23.34%, highlighting its strong profitability in the banking sector.

-Morgan Stanley (Investment Services, USA) demonstrates a healthy profit margin of 21.13%, reflecting its strong position in the investment services industry.

-recommendation: Transamerica, AIA Group, and Ping An should continue capitalizing on their strong profitability by expanding into new markets and investing in digital innovation. Meanwhile, StoneX should focus on improving operational efficiency to convert high revenue into stronger profits, and China Merchants Bank and Morgan Stanley should leverage their high margins to strengthen global competitiveness and diversify services.

```
In [25]: #Correlation Between Assets and Profitability
fin_df[['Total_Assest_in_USD_Millions', 'Profit_Margin']].corr()
```

```
Out[25]:
```

	Total_Assest_in_USD_Millions	Profit_Margin
Total_Assest_in_USD_Millions	1.000000	0.656177
Profit_Margin	0.656177	1.000000

Insight: The correlation coefficient between Total Assets and Profit Margin is 0.656, indicating a moderate to strong positive relationship. This suggests that, generally, companies with larger asset bases tend to achieve higher profit margins. It implies that asset-rich firms may be more efficient, benefit from economies of scale, or have better capacity to generate profit from their operations.

This insight could be valuable for strategic planning—firms aiming to improve profitability might focus on scaling assets strategically, ensuring those assets are well-utilized for generating income.

```
In [26]: #Rank vs Revenue Check
fin_df[['Rank', 'Revenue_in_USD_Million']].sort_values(by='Rank')
```


Out[26]:

	Rank	Revenue_in_USD_Million
0	1	245510
1	2	191509
2	3	182794
3	4	172000
4	5	153884
5	6	144589
6	7	136173
7	8	134045
8	9	129503
9	10	128011
10	11	106437
11	12	102851
12	13	97128
13	14	93753
14	15	88839
15	16	84290
16	17	82958
17	18	81632
18	19	80429
19	20	80303
20	21	78898
21	22	76984

Rank	Revenue_in_USD_Million
22	23 74074
23	24 73841
24	25 73630
25	26 67842
26	27 67605
27	28 66228
28	29 63324
29	30 62752
30	31 62579
31	32 61185
32	33 60433
33	34 59001
34	35 58840
35	36 57033
36	37 56838
37	38 55973
38	39 54139
39	40 53498
40	41 53313
41	42 52628
42	43 52047
43	44 51919

	Rank	Revenue_in_USD_Million
44	45	51516
45	46	50359
46	47	49076
47	48	48183
48	49	46788
49	50	14592

Insights: Revenue consistently declines with each successive rank, indicating a steep drop-off in scale from the top 10 companies to those ranked lower.

-Recommendation This suggests that the financial services market is top-heavy, with a few dominant firms capturing the lion's share of global revenue.

```
In [27]: # Conglomerates vs Others in Revenue and Profit
fin_df.groupby('Industry')[['Revenue_in_USD_Million', 'Net_Income_in_USD_Millions']].mean().sort_values(by='Revenue_in_USD_Mil
```

```
Out[27]:
```

	Revenue_in_USD_Million	Net_Income_in_USD_Millions
Conglomerate	245510.000000	42521.000000
Banking	88008.700000	14624.250000
Insurance	81346.565217	4054.652174
Investment Services	65850.166667	6700.500000

Industry		
Conglomerate	245510.000000	42521.000000
Banking	88008.700000	14624.250000
Insurance	81346.565217	4054.652174
Investment Services	65850.166667	6700.500000

-Insight: Conglomerates and banks generate the highest net incomes alongside their revenues, while insurance companies show the lowest profitability despite relatively high revenues

-Recommendation: This highlights a need for efficiency improvements in the insurance sector and suggests that conglomerate structures may offer better profit scalability across services.

In []: